

# GANBAT SELENGE

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**Availability:** Available all day every day except Mondays and Tuesdays; full-time during university holidays.

**Right to Work:** Eligible to work in the UK up to 20 hours/week during term-time; full-time during holidays.

## SUMMARY

Graduate Mechatronics Engineer (GPA: 4.7/5.0) and MSc student in Robotics, AI and Autonomous Systems at City St George's, University of London. Experienced in ROS 2, sensor fusion (EKF, LiDAR), and autonomous navigation. Academic research background in multi-robot systems with co-authored work submitted to IEEE PEMC. Reliable team player with experience supporting teaching, laboratory operations, and deadline-driven client work.

## EDUCATION

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| <b>City St George's, University of London</b><br><i>MSc in Robotics, AI and Autonomous Systems (with Placement)</i><br>– <b>Focus Areas:</b> Intelligent Autonomous Systems, Machine Learning, Robotic Perception, and Control.                                                                                                                                                                                                                    | London, United Kingdom<br>Sep 2026 – Sep 2028 |
| <b>University of Debrecen</b><br><i>Bachelor of Science in Mechatronics Engineering</i><br>– <b>Grade: Excellent (4.7/5.0)</b>   Total Credits: 231 ECTS<br>– <b>Specialization:</b> Mechatronic Systems (Control Theory, Robotics, and Embedded Systems)<br>– <b>Honors:</b> Consistently achieved "Excellent (5)" grades in core technical modules including Robots and Robotics Technology, Applied Automatization, and Cyber-Physical Systems. | Debrecen, Hungary<br>Sep 2022 – Feb 2026      |

## EXPERIENCE

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| <b>Autonomous &amp; Assistive Robotics Research Group (A<sup>2</sup>R)</b><br><i>Undergraduate Researcher &amp; Teaching Assistant</i><br>– Supported MSc and BSc teaching: delivered exercise classes and lab demonstrations, advised students on laboratory work, and helped organise and maintain lab equipment.<br>– Researched leader-follower coordination frameworks with a focus on odometry error correction, contributing to a co-authored conference paper. | Debrecen, Hungary<br>Feb 2025 – Feb 2026                          |
| <b>Lime Light Renhold – TechLead Department</b><br><i>Intern – Renewable Energy Management</i><br>– Gained theoretical and practical knowledge of installing, monitoring and maintaining rooftop solar panel systems, including measuring and fitting, testing, and troubleshooting and fixing issues.<br>– Covered connecting panels to an electrical system, grounding equipment and checking wiring.                                                                | Oslo, Norway<br>Jul 2025 – Sep 2025                               |
| <b>Grapecity Mongolia</b><br><i>Frontend Developer (Contract)</i><br>– Built responsive interfaces for a banking application and the "UB Parking" system, delivering to fixed client deadlines.<br>– Worked with backend teams and stakeholders to resolve API integration bugs and agree requirements, improving application load times.                                                                                                                              | Ulaanbaatar, Mongolia<br>May 2020 – Aug 2020; Jan 2021 – Feb 2021 |

## TECHNICAL SKILLS

**Robotics & Autonomy:** ROS 2 (Nav2, SLAM, Localisation, Path Planning), Sensor Fusion (EKF), PID & State-Space Control  
**Perception & ML:** OpenCV, YOLO, PyTorch/TensorFlow  
**Programming:** Python, C/C++, TypeScript/JavaScript, SQL, REST APIs, Algorithms & Data Structures, Bash  
**Embedded & Hardware:** STM32, ESP32, Raspberry Pi, Arduino, UART/SPI/I<sup>2</sup>C, LiDAR, IMU, Ultrasonic/IR Sensors  
**Software & Tools:** Linux/Ubuntu, Git, Docker, CI/CD, CMake, Testing, MATLAB, Simulink, LabVIEW, KiCad, LaTeX  
**Languages:** English (fluent), Mongolian (native)

## PUBLICATIONS

**Leader-Centric Coordination and Control of Mobile Robot Swarms for Autonomous Task Execution *IEEE PEMC 2026***  
– **S. Ganbat**, H. A. Neamah, D. Syauqy, J. N. P. Martinez, M. Al-Gburi, and P. Korondi. Co-authored paper submitted to the 22nd IEEE International Power Electronics and Motion Control Conference (PEMC 2026).

## RESEARCH & ENGINEERING PROJECTS

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| <b>Coordination and Control of Mobile Robot Swarms (BSc Thesis)</b><br>– Designed and implemented a cooperative localization and perception pipeline for multi-robot formations.<br>– Programmed Extended Kalman Filters (EKFs) to fuse LiDAR coordinate measurements with wheel odometry and IMU data.<br>– Developed a perception module using DBSCAN clustering to track follower robots in real-time.<br>– Validated the system's performance within ROS 2 Gazebo and deployed code on Raspberry Pi 4 hardware. | Feb 2026   ROS 2, Python, C++ |
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